

Sameer Hassan

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Professional Summary

Computer Science undergraduate at Bahria University (Islamabad) with a strong foundation in **C++** and core programming concepts including data structures and object-oriented programming. Comfortable with Python, SQL, and web fundamentals (HTML/CSS), and experienced building desktop applications with Qt. Completed a summer AI/ML engineering internship at Nerd Flow (Pvt) Ltd, where he shipped LLM-powered hiring features for the company's HR platform, and has built several full-stack web, mobile, and database projects to grow practical, hands-on development skills.

Experience

Nerd Flow (Pvt) Ltd — AI/ML Engineer Intern

Python · LLM APIs · Ollama · REST APIs

July 2026 — September 2026

Rawalpindi, Pakistan (Hybrid)

- Built AI-powered hiring features for **Preplify**, the company's HR management platform, including an LLM-based résumé-parsing stage that converts raw CV text into structured candidate profiles (contact details, skills, education, employment history) for downstream screening
- Designed the AI hiring system end-to-end — batched job postings, per-criterion candidate scoring against each job description (skills match, experience relevance, education fit, career stability), AI-conducted first-round interviews with transcript evaluation, and a full HR override on every automated decision; scenario analysis projected HR screening effort dropping from ~34.5 hours to ~1.75 hours per 120-applicant opening
- Benchmarked LLM options for the extraction and screening stages — self-hosted open-source models (Qwen2.5-1.5B-Instruct via Ollama, fine-tuned Qwen3-0.6B) against hosted OpenAI APIs — and documented accuracy, cost-per-résumé, and candidate-data-privacy trade-offs
- Shipped working browser prototypes of the AI hiring panel and résumé-parser flow to validate the design with stakeholders before implementation
- Researched multi-platform job posting and workflow automation (n8n), then presented technical recommendations and progress to the engineering team
- **Tech Stack:** Python, LLM APIs (OpenAI), Ollama, REST APIs, prompt design, JavaScript

Projects

Qeemat (formerly Hisaab) — Personal Finance & Receipt Manager (Solo)

Flutter · Node.js · PostgreSQL · React · Firebase

2026

sameer-hassan01.github.io

- Designed and built a full-stack, offline-first personal finance app for Pakistani users — on-device OCR receipt scanning, expense and income tracking, budgets, subscriptions, grocery price comparison and rewards — architecting three coordinated codebases solo: a **Flutter mobile app** (Riverpod, GoRouter, Dio, encrypted Hive cache), a **Node.js + Express REST API** on PostgreSQL with raw SQL via *pg*, and a **React + Vite + Tailwind admin panel**
- Built **automatic transaction capture**: an Android *NotificationListenerService* reads bank and wallet alerts and a confidence-scored parser turns them into transactions entirely on-device — message text never leaves the phone, and low-confidence captures are queued for confirmation rather than silently posted
- Hardened the app for finance-grade trust: biometric + PIN app lock with attempt lockout, *FLAG_SECURE* screenshot blocking, Hive encrypted with keys held in the Android Keystore, 15-minute access tokens with rotating refresh tokens and replay detection, EXIF stripping on uploads, and account deletion that cascades through Cloudinary and Firebase
- Modelled multi-account balances, inter-account transfers, *udhaar* (money lent and borrowed), household members with allowances and a two-level category tree over versioned SQL migrations — covered by **80+ Jest/Supertest tests**, including cross-user authorization checks, run in GitHub Actions against a real PostgreSQL instance
- Implemented an offline outbox with ordered, idempotent replay keyed on client-generated IDs (verified in airplane mode), plus idempotent scheduled jobs on Vercel Cron for renewal, budget and overdue-loan reminders
- Deployed entirely on free tiers (Vercel, Neon, Cloudinary, Firebase) at zero monthly cost; trimmed the release APK from 90.5 MB to 32.9 MB via R8, resource shrinking and per-ABI splits
- **Tech Stack:** Flutter, Dart, Riverpod, Kotlin, Node.js, Express, PostgreSQL, React, Vite, Tailwind, Firebase, Cloudinary, Google ML Kit, Jest, GitHub Actions

CineWorld — Cinema Ticketing & Management System (DBMS Semester Project) 2026
PHP · MySQL · HTML/CSS/JS *github.com/sameer-hassan01/cinemaManagement-DBMS*

- Built a full-stack cinema booking and management web application end-to-end: user registration with role selection, movie browsing, day/time/seat selection, mock payment flow, and printable PDF tickets via FPDF
- Designed and normalised a 9-table MySQL schema covering users, movies, screenings, screens, seats, shows, bookings, contact messages, and upcoming releases
- Implemented a role-guarded admin dashboard for adding, editing, and deleting movies; viewing all bookings; and reading customer messages from the contact form
- Wrote every database query using *mysqli* prepared statements to eliminate SQL injection vectors; hashed passwords with PHP's *password_hash* (bcrypt)
- Integrated Cloudinary's free tier for movie poster hosting
- **Tech Stack:** PHP, MySQL, HTML, CSS, JavaScript, Cloudinary, FPDF, XAMPP

FoodBook — Community Food Review Platform (Team Project) 2025
JavaScript · REST APIs · Git *Collaborative MVP*

- Worked with another developer to build the MVP of **FoodBook**, a community-driven food review platform
- Contributed feature slices including review submission, rating logic, and search, and wrote unit tests for the modules I authored
- Took part in bug-fixing during the beta phase, supported deployment, and participated in code reviews
- **Tech Stack:** JavaScript, REST APIs, Unit Testing, Git

University Coursework Projects 2023 — Present
C++ · Python · Qt · SQL · scikit-learn *Bahria University, Islamabad*

- Built multiple coursework projects across Data Structures, OOP, Database Systems, and Machine Learning — including an intrusion-detection system project and ML pipelines in Python with pandas and scikit-learn
- Worked in small student teams using Git for version control, with end-to-end ML pipelines covering feature engineering, cross-validation, and model evaluation
- **Tech Stack:** C++, Python, Qt, SQL, scikit-learn, Git

Education

Bahria University August 2023 — Present
BS in Computer Science (specialisation: machine learning & data science) *Islamabad, Pakistan*

- Relevant coursework: Machine Learning, Deep Learning, Data Mining, Probability & Statistics, Data Structures & Algorithms, Database Systems, Operating Systems, Computer Networks
- Built end-to-end ML pipelines in Python (pandas, NumPy, scikit-learn) with feature engineering, cross-validation, and hyperparameter tuning

Islamabad College for Advanced Studies (ICAS) August 2021 — June 2023
A Levels in Mathematics, Physics, & Computer Science *Islamabad, Pakistan*

- Capstone project: a dynamic web application built with JavaScript and Flask

Fazaia Education System School June 2020 — June 2021
O Levels — Grade A in Computer Science *Islamabad, Pakistan*

- Foundational programming and problem-solving with hands-on projects in Python and basic web development

Technical Skills

Languages: C++ (proficient), Python, SQL, Visual Basic, HTML / CSS

Tools & Frameworks: Qt (Qt Creator & Qt Designer), Git & GitHub

AI / ML: LLM APIs (OpenAI), prompt design, local models via Ollama, pandas, scikit-learn

Concepts: Data Structures, Object-Oriented Programming, Basic Database Design, RAG & agent workflows (basic)